PAKISTAN

HQ-9/P Chinese-made long-range SAM (100-200 km), less advanced than S-400

OP SINDOOR: Claimed to have downed 25 Indian drones; one likely lost to an Indian missile strike in Karachi



FM-90 Chinese short-range SAM (15 km); limited range, rapid response



OP SINDOOR: Tasked to protect frontline air bases like Nur Khan, ultimately failed to keep away Indian missiles on May 10



EYE ON INDIA
A TPS-77 MRR ground-based radar in Pakistan

"ABC" SYSTEM Tactical air defence linking ground-based radars, J-10C/ JF-17 jets, AWACS; enhances real-time targeting

OP SINDOOR: Claimed to have guided the PL-15 missiles that downed Indian aircraft

PAKISTAN'S INVESTMENT IN OFFENSIVE WEAPONS LAYS BARE ITS WEAK AIR DEFENCE CAPABILITIES, SAYS A DEFENCE EXPERT. DEVELOPMENT OF MISSILES LIKE BRAHMOS AND AKASH, TIMELY PROCUREMENT OF THE S-400 GAVE INDIA A BIG EDGE IN NON-CONTACT WARFARE

missiles and HAMMER precision-guided munition, is India's most advanced jet, but its small fleet size limits impact.

On the other side, Pakistan's Chinese J-10C fighters, armed with PL-15E BVR missiles, and equipped with dual-pulse motors and AESA seekers, mounted a challenge for the IAF. The J-10CE's AESA radar and its integration with Saab 2000 Erieye AEWC enables long-range engagements, while the JF-17's KLJ-7A radar is less capable but still effective with networked support.

Operation Sindoor drew global attention to a possible aerial duel between the Rafale and J-10C fighters—both 4.5-generation multirole fighters. "Rafale is a proven platform, as it has been used in combat in Afghanistan, Libya, Mali, Iraq and Syria, while J-10 C has not seen any combat," says a key IAF official.

Military aviation experts claim that Pakistan's J-10C, manufactured by the Chengdu Aircraft Industry Group, lags behind the Rafale, made by the French firm Dassault, in armament. The J-10C has 11 hardpoints (mounting points to carry weapons) and a six-tonne load capacity. The Rafale has 14 hardpoints and a nine-tonne capacity, including nuclear-capable munitions. The Rafale's 24-tonne take-off weight exceeds the J-10C's 19 tonnes, offering greater payload flexibility, though the J-10C reaches a higher 18,000-metre ceiling, compared to the Rafale's 16,000 metres. The J-10C excels in high-altitude performance; the Rafale dominates in versatility and range. Experts say that the Rafale holds a slight edge in terms of sensor fusion and missile performance. However, uncertainty persists over Pakistan's unverified claims of downing Indian jets, including Rafales.

"Public evidence does not confirm that a Rafale was downed by a J-10C-fired PL-15E missile. However, the loss of at least one Rafale demands an urgent review of technical and tactical vulnerabilities," notes O'Donnell. He adds that Pakistan and China have valuable combat data about western aircraft, aiding their fighter modernisation, while Pakistan's swift acquisition of Chinese platforms outpaces India's sluggish defence procurement, exacerbating the IAF's declining squadron strength.